# Multiple Components

# Send data to Component

- Data to a component can be sent as an attribute in the template

```
<app-display   [name]= “myName”>  </app-display>
```

- Child component should have input property to use this value
- Input property is provided with @Input() decorator

```
export class  DisplayComponent {
 @Input()  name:string;
}
```

# Output binding

- Output binding done through event binding.
- the attribute is of type EventEmitter
- child element can emit an  event and bind that event to output

```
export class DisplayComponent {
  @Output() change = new EventEmitter();

  clicked(){
     this.change.emit();
  }
}
```

- Usage:

```
<app-display  (change)='dataChanged()'></app-display>
```

function in parent

# Passing data in Output

- emit() method can have some data as argument
- This is linked  to $event object of the output event.
- **$event** - Is an expression exposed in event bindings by Angular, which has the value of the event's payload
- $event represents the data sent by emit()
- It is not same $event in normal javascript
- Example: (child component)

```
export class  DisplayComponent {
@Output() change = new EventEmitter();
 email:string;

 send(){
        this.change.emit(this.email);
  }
}
```

# Passing data in Output

- Example: (child component  template)

```
<input [(email)] />
<button (click)="send()">Click</button>
```

- Example: (parent component  template)

```
<app-display  (change)='dataChanged($event)'></app-display>
```

# Passing data in Output

- Example: (parent component )

```
export class AppComponent {

  dataChanged(email){
    console.log("Email of person now is "+ email);
  }
}
```

# ng-content

- Ng-content object can be created in child object to put content coming from parent
- Use select=” ” as a CSS selector
- Parent will create such elements while using child components
- Selector is matched to place the content of the parent

# ng-content - Example

Child Component

```
<div>
  <div >
    <ng-content  select='.heading'></ng-content>
  </div>
  <div>
    <ng-content  select=‘.body'></ng-content>
  </div>
</div>
```

# ng-content - Example

Parent Component

```
<app-child>
   <div class='heading'>This is Heading</div>
   <div class="body">
      <h1>line1</h1>
      <h2>line2</h2>
   </div>
</app-child>
```

# ng-content - Example

Child  Component with content sent by parent

```
<div>
  <div  >
      <div  class="heading" >This is Heading</div>
  </div>
  <div>
     <div  class="body">
        <h1>line1</h1>
        <h2>line2</h2>
     </div>
</div>
```

# @ViewChild / @ViewChildren

- This decorator used to get reference of any dom element or child component
- Using the reference variable, the element properties or methods can be accessed and manipulated
- @ViewChild refers the first matching element
- @ViewChildren refers all matching elements as QueryList array

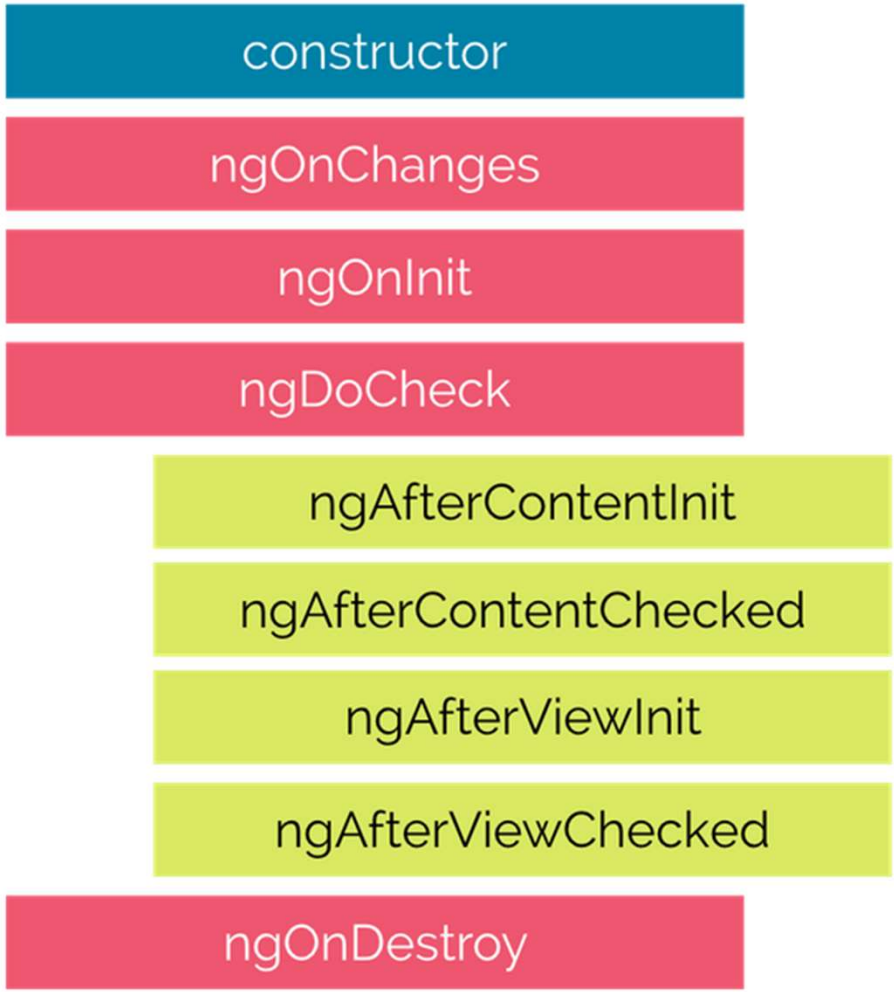
# @ContentChild / @ContentChildren

- Used for accessing content received by paren as ng-content

# Component lifecycle

- A component in Angular has a life-cycle, a number of different phases it goes through from birth to death.
- We can hook into those different phases to get some pretty fine grained control of our application.
- To do this we add some specific methods to our component class which get called during each of these life-cycle phases, we call those methods hooks.
- We can create methods to execute at every phase

# Component lifecycle phases



# Component lifecycle

**constructor**

This is invoked when Angular creates a component or directive by calling new on the class.

**ngOnChanges**

Invoked every time there is a change in one of th input properties of the component.

**ngOnInit**

Invoked when given component has been initialized.

This hook is only called once after the first ngOnChanges

**ngDoCheck**

Invoked when the change detector of the given component is invoked. It allows us to implement our own change detection algorithm for the given component.

**ngOnDestroy**

This method will be invoked just before Angular destroys the component.

Use this hook to unsubscribe observables and detach event handlers to avoid memory leaks.

# Component lifecycle

**ngAfterContentInit**

Invoked after Angular performs any content projection into the component's view .

**ngAfterContentChecked**

Invoked each time the content of the given component has been checked by the change detection mechanism of Angular.

**ngAfterViewInit**

Invoked when the component's view has been fully initialized.

**ngAfterViewChecked**

Invoked each time the view of the given component has been checked by the change detection mechanism of Angular.

# Change Detection Strategies

- Change detection in Angular is executed when:
  - There are updates to Angular State variable
  - Events are invoked inside Angular Component
  - @Input Values are updated for the Components

- Change Detection Strategies are the methods by which the updates to the component is tracked and component is triggered to Re-render
- There are majorly 2 Change Detection Strategies
  - Default Strategy ← this is default
  - onPush Strategy

# Implementing onPush strategy

```
@Component({
  selector: "child-component",
  template: `
    <div>
      <h2>{{ this.executeFunction() }}</h2>
    </div>
  `,
  changeDetection: ChangeDetectionStrategy.OnPush
})
```

# Dynamic Components

- Dynamic means, that the components location in the application is not defined at buildtime.
- That means, that it is not used in any angular template.
- Instead, the component is instantiated and placed in the application at runtime
- To load dynamic component, first we need to create a container and at runtime get ViewContainerRef through ViewChild

# Angular Forms

# Bootstrap for Angular

- For using bootstrap in Angular, module can be installed

  ng add @ng-bootstrap/schematics

- bootstrap entry will be added to package.json

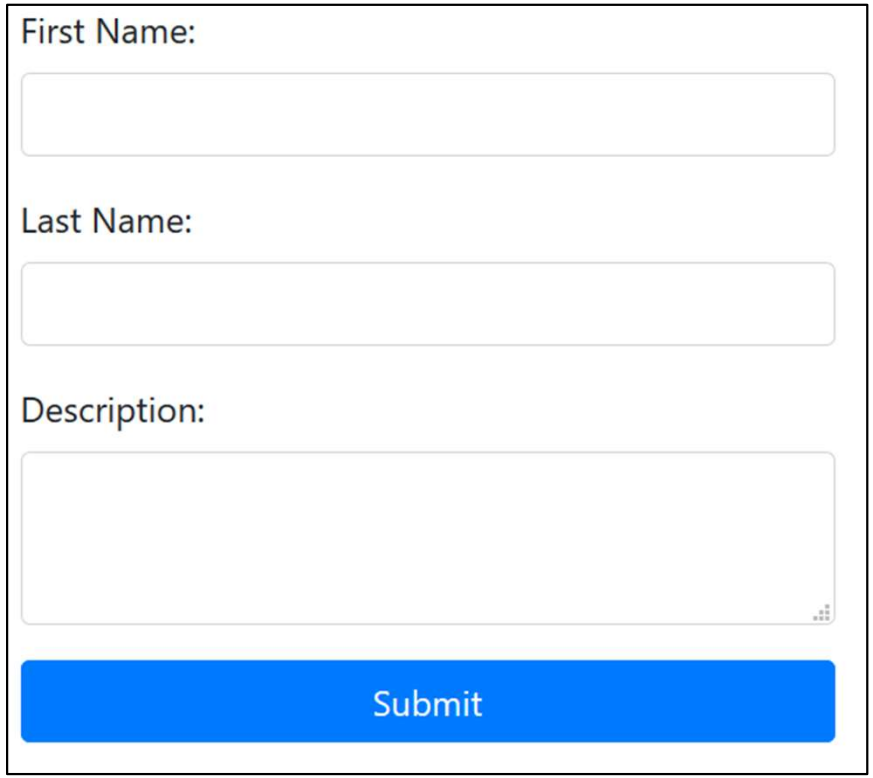
- Use bootstrap classes in templates

```
<div class='form-group' style="width:400px">
    <label for='input1'>Enter id</label>
    <input class='form-control'  id='input1'/>
    <label for="input2">Enter Name</label>
    <input class='form-control'  id='input2'/>
  <div>
```

# Bootstrap for Angular

Normal Form

Form using Bootstrap



# Angular Form Control

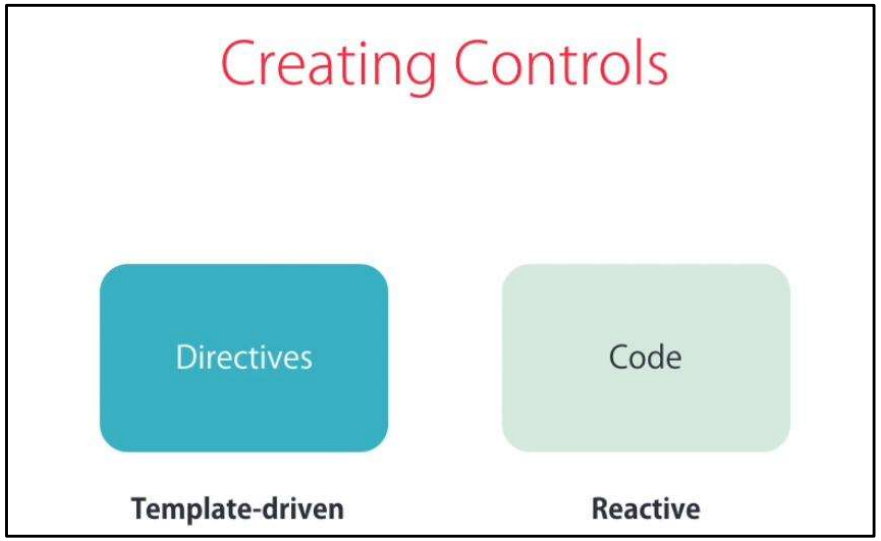
- Following classes used for Form Control
  - FormControl: Associated with each input field
  - FormGroup: associated with multiple input fields
- FormControl has properties:
  - value — value of the input field
  - touched — boolean : true if the field is touched
  - untouched — boolean : true if the field is untouched
  - dirty — boolean : true if the value is changed
  - pristine — boolean : true if the value is not changed
  - valid — boolean : true if field passes all validations
  - errors — provides all the validation errors
- FormGroup has all above properties with respect the group

# Angular Form Control

- For form control and validations  add  FormControl object to each input field and FormGroup object to the form

# Creating Controls

- **Template Driven Forms**
  - Controls created by applying Directives
  - Angular creates the control objects implicitly
  - Good for simple forms with basic validations like
    - required
    - range of values etc
- **Reactive Forms**
  - Controls created using custom code
  - Give more control on validation logic
  - Good for complex forms
  - Also called as Model driven Forms



# Template Driven Forms

# Template Driven Approach

- Teplate Driven forms use ngModel directive

- Input fields should be associated with :-
  - ngModel directive
  - name  attribute
  - Template variable

# ngModel

- With this directive angular creates all associated controls for the input field
- name attribute is required to associate the ngModel

```
<input   ngModel  name='courseName'  />
```

- To refer the input field and its associated control properties in the template, we can create template variable for ngModel

```
<input  ngModel name='courseName'   #course="ngModel"/>
```

- Template variable and name attribute are two different things

# More on ngModel

- Value of the name attribute is the name of the property to represent the value of the field
- Template variable is an object of ngModel and used to refer the field for validations etc

```
<form  #form="ngForm">
   <label for="fn">First Name</label>
   <input id='fn' type="text" ngModel name='fristName'  #nameField='ngModel' />
   <label for="age">Age</label>
   <input id='age' type='text' ngModel name='age'   #ageField='ngModel' />

   <div>{{form.value|json}}</div>
</form>
```

First Name

Sudhir

Age

35

{ "fristName": "Sudhir", "age": "35" }

# Validation

- Basic validation supported in HTML5 can be done with ngModel
- ngModel object has properties like valid, touched, dirty, pristine, invalid, errors etc
- Example:

```
<div class='holder'>
  <div class='form-group'>
    <label for="name">Name</label>
    <input id='name' ngModel name='personName' required minlength='5' maxlength='20'
          class="form-control" #name='ngModel'  />
    <div class='alert alert-danger' *ngIf='name.touched && name.invalid'>
      <div *ngIf='name.errors.required'>Name required</div>
      <div *ngIf='name.errors.minlength'>
        Minimum {{name.errors.minlength.requiredLength}} characters required
      </div>
    </div>
  </div>
```

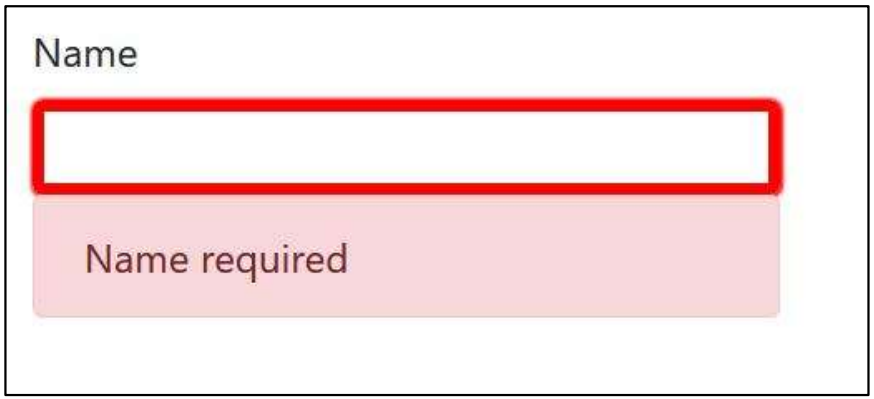
# ngModel style classes

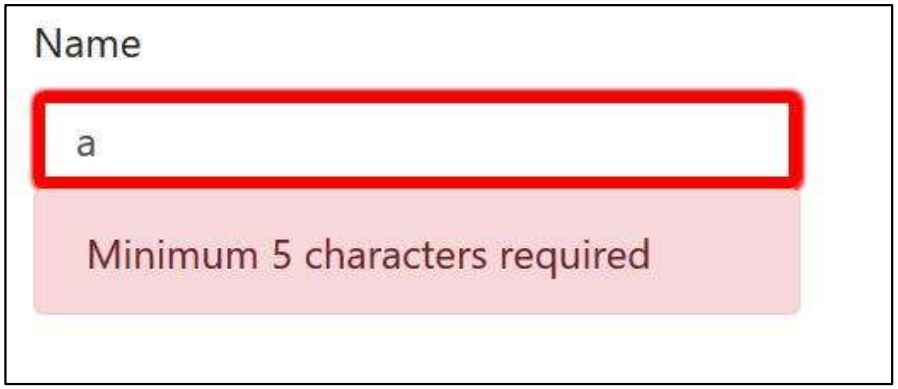
- The *ngModel* directive doesn't just track state; it updates the control with special Angular CSS classes that reflect the state
- You can leverage those class names to change the appearance of the control

| State | Class if true | Class if false |
|---|---|---|
| The control has been visited. | ng-touched | ng-untouched |
| The control's value has changed. | ng-dirty | ng-pristine |
| The control's value is valid. | ng-valid | ng-invalid |

# ngModel style classes

```
.ng-touched.ng-invalid {
   border: solid 2px red;
 }
```



# ngForm

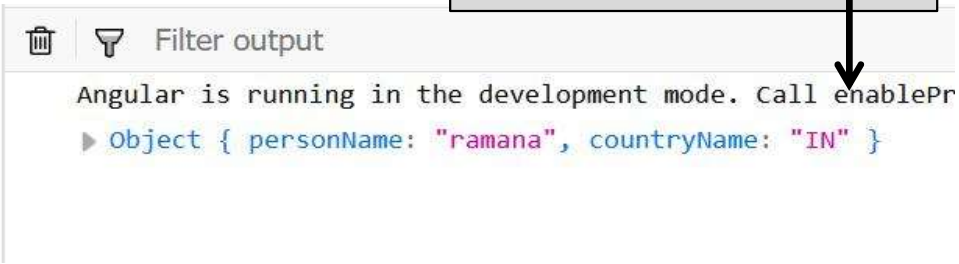
- ngForm directive is automatically applied on <form> tag
- This directive has output property ngSubmit
- ngForm creates a FormGroup object for the form
- We can access all the properties like touched, dirty, valid, invalid etc using a template name for ngForm
- value property represents an object holding values of all input elements in the group

# ngForm - Example

```
<form #userForm="ngForm" (submit)=submitForm(userForm.value)>
  <div>
    <label>Name</label>
    <input ngModel name='personName' class="form-control"  #name='ngModel' />
    <label>Country</label>
    <select id='country' ngModel name='countryName' class="form-control" #country='ngModel'>
      <option *ngFor="let c of countries" [value]='c.id'>{{c.name}}</option>
    </select>
    <button type='submit' class='btn btn-primary'>Submit</button>
  </div>
</form>
```

```
countries = [
   { id: "IN", name: 'India' },
   { id: "US", name: 'U S A' },
   { id: "FR", name: 'France' }
  ];

  submitForm(data) {
   console.log(data);
  }
```



# Disabling Submit for invalid Form

- As a whole it is possible to check errors for the entire form using ngForm directive
- All properties of ngModel are available in ngForm
- For example, we can disable submit button if any of the input fields are invalid

```
<form #userForm="ngForm" (submit)=submitForm(userForm.value)>

  <button type='submit' class='btn btn-primary'
            [disabled]='userForm.invalid' >Submit</button>
</form>
```

# ngModelGroup

- This directive can only be used as a child of NgForm (or in other words, within <form> tags)
- Use this directive  to create a sub-group within a form
- Helps to validate a sub-group of the form separately from the rest of the form
- Pass in the name you'd like this sub-group to have and it will become the key for the sub-group in the form's full value
- You can also export the directive into a local template

  ex: #myGroup="ngModelGroup"

# ngModelGroup - Example

```
<form #form="ngForm" (ngSubmit)="onSubmit(form)">
    <p *ngIf="nameCtrl.invalid">Name is invalid.</p>

    <div ngModelGroup="name" #nameCtrl="ngModelGroup">
      <input name="first"  minlength="2">
      <input name="last"  required>
    </div>
</form>
```

In the code:

```
onSubmit(form){
   console.log(form.value.name.first);
   console.log(form.value.name.last);
}
```

In the template :

```
<div *ngIf="!nameCtrl.valid">
    error messages for the name group
</div>
```

# Using checkbox

```
<div class='checkbox'>
  <label>
     <input type="checkbox" ngModel name="isSubscribed">
  </label>
</div>
```

# Using radio group

```
<div class='radio' *ngFor="let method of contactMethods">
  <label>
    <input type="radio"  ngModel
           name="contactMethod"   [value]="method.id">
    {{method.name}}
  </label>
</div>
```

Data in the component:

```
contactMethods = [    { id: 1, name: 'email' },
                      { id: 2, name: 'phone'},
                      { id: 3, name: 'Languages' },
                  ];
```

# Using dropdown list

```
<label for="country">Country</label>
 <select id='country' required   name='country'
          class="form-control" ngModel  #country='ngModel'>
     <option ></option>
     <option *ngFor="let c of countries" [value]='c.id'>{{c.name}}</option>
 </select>

 <div class='alert alert-danger' *ngIf='country.touched && country.invalid'>
      Select one country
 </div>
```

Data in the component:

```
countries = [
        { id: "IN", name: 'India' },
        { id: "US", name: 'U S A' },
        { id: "FR", name: 'France' }
 ];
```

# Reactive Forms

# Reactive Forms

- Reactive forms provide a model-driven approach to handling form inputs
- Instead of defining the form in the template, the structure of the form is defined in the code
- In Template Driven Forms, the FormGroup and FormControl objects are automatically created with ngForm directive
- In Reactive forms, these objects have to be created in the code and linked to the input form of the template
- To use reactive forms, we need to import the ReactiveFormsModule into our parent module

# Registering ReactiveFormsModule

- import ReactiveFormsModule from the @angular/forms package and add it to your NgModule's imports array

```
import { ReactiveFormsModule } from '@angular/forms';

@NgModule({
  imports: [
    // other imports ...
    ReactiveFormsModule
  ],
})
export class AppModule { }
```

# Simple Form with one field

- import ReactiveFormsModule from the @angular/forms package and add it to your NgModule's imports array

```
import { FormControl } from '@angular/forms';

@Component({
  selector: 'name-app',
  templateUrl: './name.component.html',
  styleUrls: ['./name.component.css']
})
export class NameComponent {
  name = new FormControl('Ramana');
}
```

```
<label>  Name:</label>
<input type="text" formControlName="name">
```

# FormControl

- Some of the properties / methods of FormControl
  - value
  - setValue() -- setting the value directly is not possible
  - disable()
  - disabled -- boolean
  - enable()
  - enabled -- boolean
  - valid
  - dirty
  - prestine
  - …… etc

# Grouping Controls

- Just as a form control instance gives control over a single input field, a form group instance tracks the form state of a group of form control instances (for example, a form).
- Each control in a form group instance is tracked by name when creating the form group

- Creating FormGroup :

```
import { FormGroup, FormControl } from '@angular/forms';

@Component({
  selector: ‘app-user’,
  - - - -
})
export class UserComponent {
     userForm = new FormGroup({
          firstName: new FormControl(‘ '),
          lastName: new FormControl(‘ '),
      });
}
```

# Associating FormGroup control to View

- A form group tracks the status and changes for each of its controls
- Use formGroup attribute at form level
- Use **formControlName** for each element
- The model for the group is maintained from its members

```
<form [formGroup]=“userForm">

  <label  for=‘first> First Name: </label>
  <input  id= ‘first’ type="text" formControlName="firstName">

  <label  for=‘last> Last Name: </label>
  <input  id=‘last’ type="text" formControlName="lastName">

</form>
```

# Validation with Reactive Forms

- HTML5 built-in attributes can be used for native validation, including required, minlength, and maxlength

```
<form [formGroup]=“userForm">
<div class='form-group'>
<label> First Name:</label>
<input  type="text" formControlName="firstName"  required >
</div>

<div class="alert alert-danger"
      *ngIf=“userForm.get('firstName').touched && userForm.get('firstName').invalid">
   First Name is Required
</div>

<div class='form-group'>
<label for="last"> Last Name:</label>
<input id='last' type="text" class='form-control' formControlName="lastName" >
</div>
</form>
```

# Validators class

- Reactive forms include a set of validator functions which are static methods of Validators class
- Validations can be defined in the code by providing Validator functions in the FormControl object
- We can supply one function or array of functions

- Import the Validators class from the @angular/forms package

# Validators class

```
import { Validators } from '@angular/forms';
…….

export class Form2Component {
   userForm: FormGroup = new FormGroup(
    {
        firstName: new FormControl(‘ ‘, [
            Validators.required,
            Validators.minLength(5)
        ]),

       lastName: new FormControl(' ‘ ,  [
             Validators.required,
             Validators.minLength(5)
       ]),
    })
}
```

# Validators

```
form [formGroup]=“userForm">
  <div class='form-group'>
    <label> First Name:</label>
    <input  type="text" class='form-control' formControlName="firstName">
  </div>
  <div class="alert alert-danger"
      *ngIf=“userForm.get('firstName').touched && userForm.get('firstName').invalid">

   <div *ngIf=“userForm.get('firstName').errors.required">
     First Name is required
    </div>
    <div *ngIf=“userForm.get('firstName').errors.minlength">
     First Name should be minimum {{userForm.get('firstName').errors.minlength.requiredLength}} characters
    </div>
  </div>
  . . . . . .
</form>
```

# Custom Validators

- Custom validator is a simple function which takes one input parameter of type AbstractControl
- If the validation fails, it returns an object, which contains a key-value pair
- **Key** is the name of the error and the value is always **Boolean true**.
- If the validation does not fail, it returns **null**

```
function ageRangeValidator(control: AbstractControl) {
    if (isNaN(control.value) ){
        return { ‘invalidValue': true };
    }

    If( control.value < 18 || control.value > 45)) {
        return { ‘invalidRange': true };
    }
    return null;
}
```

# Custom Validators – How to use

```
loginForm = new FormGroup({
        email: new FormControl(null, [Validators.required]),
        password: new FormControl(null, [Validators.required]),
        age: new FormControl(null, [ageRangeValidator])
});
```

# Custom Validators with parameters

- Passing parameters to validator function is not possible
- But it can used as a closure by defining it inside another function which takes parameters and return validator function

```
function ageRangeValidator(min: number, max: number): ValidatorFn {
        return (control: AbstractControl): { [key: string]: boolean } | null => {
            if (control.value !== undefined && (isNaN(control.value) {
                return { 'invalidValue': true };
            }

            If( control.value < 18 || control.value > 45)) {
                return { 'invalidRange': true };
            }
            return null;
        };
}
```

```
loginForm = new FormGroup({
        .......
        age: new FormControl(null, [ageRangeValidator(25, 65)])
});
```

# Form Builder

- FormBuilder is the helper API to build forms in Angular.
- It provides shortcuts to create the instance of the FormControl, FormGroup or FormArray
- FormBuilder is injected into constructor

```
constructor(private formBuilder:FormBuilder){}
……————

this.contactForm = this.formBuilder.group({
        name: ['', [Validators.required, Validators.minLength(10)]],
        email: ['', [Validators.required, Validators.email]],
        gender: ['', [Validators.required]],
        country: ['', [Validators.required]],
        address: this.formBuilder.group({
                city: ['', [Validators.required]],
                street: ['', [Validators.required]],
                pincode: ['', [Validators.required]],
         })
});
```

# Form Array

- FormArray is similar to FormGroup and it is used as an array that wraps around an arbitrary amount of FormControl, FormGroup or even other FormArray instances
- With **FormArray** we can add new form fields dynamically

```
angForm = new FormGroup({
        names: new FormArray([
            new FormControl('', Validators.required),
            new FormControl('', Validators.required), ])
 });
```

# Routing and Navigation

# Routing in S P A

- Navigation in Single page Application is different from traditional web applications
- An SPA is a web application that provides a user experience similar to a desktop application
- In an SPA, all communication with a back end occurs behind the scenes
- When a user navigates from one page to another, the page is updated dynamically without reload, even if the URL changes

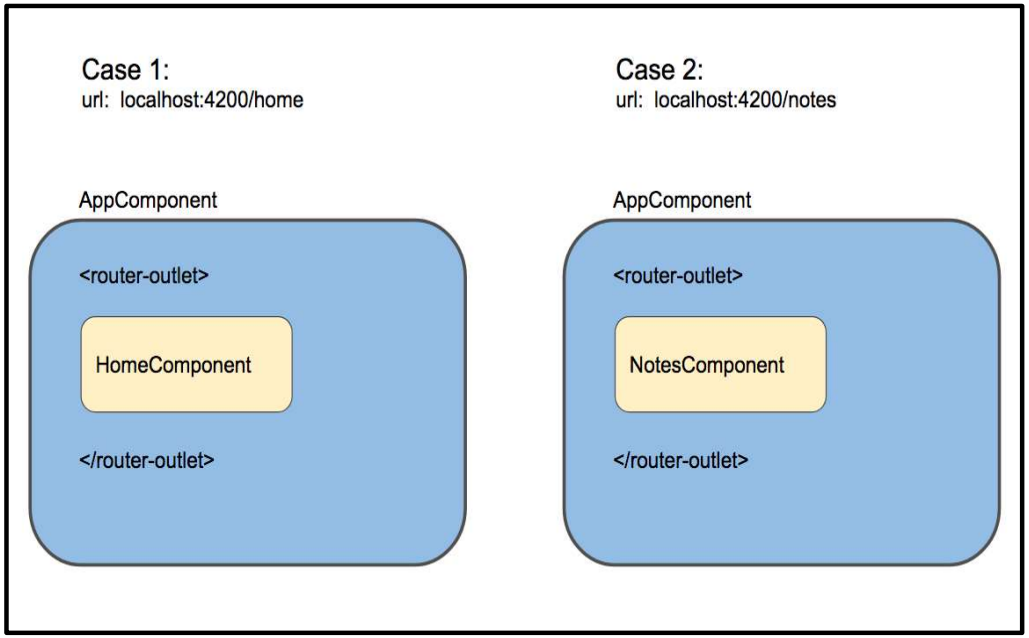
# Angular Router

- It's a JavaScript router implementation that's designed to work with Angular and is packaged as **@angular/router**
- Angular Router takes care of the duties of a JavaScript router
- It activates all required Angular components to compose a page when a user navigates to a certain URL
- It lets users navigate from one page to another without page reload
- It updates the browser's history so the user can use the *back* and *forward* buttons when navigating back and forth between pages.

# Routing in Angular

- Angular Router module helps in navigation in a single page

- Steps in Angular Routing:
  - Configure Routes - mapping of route with component
  - Add Router outlet - to display corresponding component
  - Add links for different routes



# Routing Step 1

```
import { NgModule } from '@angular/core';
import { Routes, RouterModule } from '@angular/router';
import { CoreComponent } from './core/core.component';
const routes: Routes =
 [
      { path:'home', component:HomeComponent },
      { path:'members', component:MembersComponent }
];

@NgModule({
declarations: [  . . . .  ],
imports: [
   BrowserModule,
   RouterModule.forRoot(routes)
 ],
 providers: [],
 bootstrap: [AppComponent]
})
export class AppModule { }
```

configuring Routes

import RouterModule for top level

# Routing Step 2 & 3

- While defining links we should use routerLink directive (defined RouterModule) instead of href
- href loads the whole page including all JS files which is a waste of time
- routerLink loads only the required component

```
<h3>Routing</h3>
<a routerLink='/home'>Home</a>
<a routerLink='/members'>Members</a>
<br><br>

<router-outlet></router-outlet>
```

**Links for the routes**

**router-outlet for displaying component based on the clicked link**

# Wildcard routes

- Default link for ‘not found’ page can defined with wild card
- wild card route should be the last one in the list
- Wild card route will match all the routerLink  references

```
const routes: Routes =
 [
        { path:'home', component:HomeComponent },
        { path:'members', component:MembersComponent },
        { path:’**’ , component:NotFoundComponent }
];
```

# Route Parameters

- Suppose we have an application that displays a product list and when the user clicks on a product in the list, we want to display a page showing the detailed information about that product

- To do this you must:
  - add a route parameter ID
  - link the route to the parameter
  - add the service that reads the parameter

- Route parameters added to the route with ‘:’ symbol

# Declaring Route Parameter

```
const routes: Routes = [
  { path: 'product-list', component: ProductList Component},
  { path: 'product-details/:productId', component: ProductDetailsComponent }
];
```

- **:productId** in the path of the product-details route is route parameter in the path
- For example, to see the product details page for product with ID 5, we must use the following URL:

  localhost:3000/product-details/5

# Linking to Route with Parameter

- The **routerLink** directive passes an array which specifies the path and the route parameter
- **Do not use href**. Every time you click, it downloads the whole page
- routerLink loads the resources locally through javascript
- Use routerLink attribute directive for link with no parameters
- Use routerLink property attribute if link has parameters
  - First element of the array is path
  - Second clement onwards provide values for route parameter

```
<a  routerLink='products'> Get All Products </a>

<a *ngFor="let product of products"
  [routerLink]="[ 'product-details', product.id]">
  {{ product.name }}
</a>
```

# Defining Route programmatically

- Use Router service of angular to navigate to a route
- Router is added in the component through dependency injection

```
Import   {Router}  from ‘@angular/router’;
// other code

constructor(private router: Router) { }
// other code

goToProductDetails(id) {
  this.router.navigate(['/product-details', id]);
}
```

# Reading Route parameters

- The ProductDetailsComponent must read the parameter, then load the product based on the ID given in the parameter
- The ActivatedRoute service provides a paramMap Observable to which we can subscribe to get the route parameters
- ActivatedRoute added to the component through dependency injection

```
import { ActivatedRoute } from '@angular/router';

export class ProductDetailsComponent implements OnInit{
  selectedProduct:Product;
    constructor(private route: ActivatedRoute) { }

  ngOnInit() {
  this.route.paramMap.subscribe(params => {
      let id = +params.get('productId');
      this.selectedProduct  = .... // get theproduct using id
    });
  }
```

**Using unary + to convert into number**

# Why ParamMap Observable

- If navigation happens from component A to Component B, Angular destroys Component A and creates Component B
- If navigation happens from Component A to Component A, it is not destroyed and onInit() method is not invoked
- In such case getting route parameter is a problem

- If we subscribe to paramMap observable, our callback method is called for each navigation

- If same component navigation is not required, we can use snapshot.paramMap which is just object

```
ngOnInit() {
  let  id = +this.route.snapshot.paramMap.get('productId')
  this.selectedProduct  = .... // get theproduct using id;
  }
```

# Child routes

- When some routes may only be accessible and viewed within other routes it may be appropriate to create them as child routes.
- localhost:4200/product-details/3/overview  will load OverView component

```
export const routes: Routes = [
  { path: '', redirectTo: 'product-list', pathMatch: 'full' },
  { path: 'product-list', component: ProductList },
  { path: 'product-details/:id', component: ProductDetails,
    children: [
      { path: '', redirectTo: 'overview', pathMatch: 'full' },
      { path: 'overview', component: Overview },
      { path: 'specs', component: Specs }
    ]
  }
];
```

# Child routes

- Where would the components for these child routes be displayed?
- Just like we had a <router-outlet></router-outlet> for the root application component, we would have a router outlet inside the ProductDetails component

```
@Component({
  selector: 'product-details',
  template: `
    <p>Product Details: {{id}}</p>
    <nav>
      <a [routerLink]="['overview']">Overview</a>
      <a [routerLink]="['specs']">Technical Specs</a>
    </nav>
    <router-outlet></router-outlet>
})
export default class ProductDetails implements OnInit {
  id: number;
  constructor(private route: ActivatedRoute) {}
  ngOnInit() {
    this.route.params.subscribe(params => {
      this.id = +params['id']; // (+) converts string 'id' to a number
    });
  }
```

# Accessing parent route parameters

- The child route component can access the parent route's parameters as follows component

```
export default class Overview {
  parentRouteId: number;
  private sub: any;
  constructor(private router: Router,  private route: ActivatedRoute) {}

  ngOnInit() {
    // Get parent ActivatedRoute of this route.
    this.sub = this.router.routerState.parent(this.route)
      .params.subscribe(params => {
        this.parentRouteId = +params["id"];
      });
  }

  ngOnDestroy() {
    this.sub.unsubscribe();
  }
}
```

# Lazy Loading

- Lazy loading makes it possible to load a module when it is required
- Steps:
  - Create a module with the command

    ng g m admin –route admin –m app
  - This will create AdminModule along with its own routermodule and also creates path admin
  - It also adds route for lazy loading path for this module

```
const routes: Routes = [{ path: 'lazy', loadChildren: () =>
import('./lazy/lazy.module').then(m => m.LazyModule) }];
```

  - Do not include this module in AppModule imports

# AuthGuard

- The auth guard is an angular route guard that's used to prevent unauthenticated users from accessing restricted routes
- it does this by implementing the CanActivate interface which allows the guard to decide if a route can be activated with the canActivate() method
- If the method returns true the route is activated (allowed to proceed), otherwise if the method returns false the route is blocked.
- The auth guard uses the authentication service to check if the user is logged in

# AuthGuard

```
export class RouteGuardService extends AuthGuard {

  canActivate() {
    let isAuthenticated = super.canActivate();
    if (!isAuthenticated)
      return false;

    if (this.authService.currentUser.admin)
      return true;

    this.router.navigate(['/no-access']);
    return false;
  }
}
```

# Router Guards to protected routes

```
const routes: Routes = [

{path:'emplist',component:EmpListComponent,canActivate:[Route
GuardService]},

  {path:'home',component:HomeComponent},

];
```

# Dynamic Routing

- Limitations of static routes

```
const routes: Routes =
        { path: ‘users', redirectTo: '/first-component'}  ]
```

- This shows redirect route cannot be determined at runtime
- however, insert a Route Guard into the route that is invoked when the path matches

```
{
  path: "users",
  canActivate: [RedirectService],   component: UsersComponent
},
```